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Project 3 Final Submission

Navigating Files

- /final/ - contains this report
- /finalBatches/ - contains executables and source code for all implementations as well as batch files.
- /blas/ - contains source code for cpu implementations. Has two makefiles, one for blas and one for openmp
- Gpu – contains source code and makefiles for gpu native
- gpuBlas – contains source code and makefiles for gpu with cuBLAS
- plotting – contains plots and code for plotting loss vs epochs

Creating files – go into the blas, gpu, or gpuBlas and type “make” then run the executable. Can also go into the finalBatches file and queue any of the batchfiles in there.

BLAS Implementation – OpenBLAS

Discussion

The main short comings of this project where the back and forth between the GPU which I believe causes most of the slow down. Would like to experiment with having most of the code run on the GPU. Experimenting more with batch tile sizes could also be a source of improvement as well as more efficient allocation and deallocation of memory. There is also improvements to be had to implement more cuBLAS functions, I only translated the main matrix multiplication functions. Unoptimized I/O could also be a bottleneck.

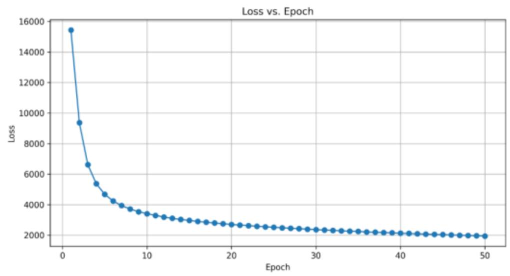
Table 1

Version	Processor	Accuracy	Grind Rate	Training Time (s)	TPB/CPU Cores
GPU Native	A100	0.94	85,959	34.9	256
GPU cuBLAS	A100	0.94	121,000	24.7	NA
CPU Native	NA	0.94	37,641	79.7	16
CPU BLAS	NA	0.94	9,933	302.3	1

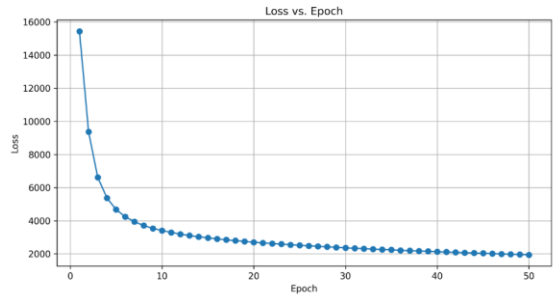
(LOSS CURVES ON NEXT PAGE)

Loss Curves

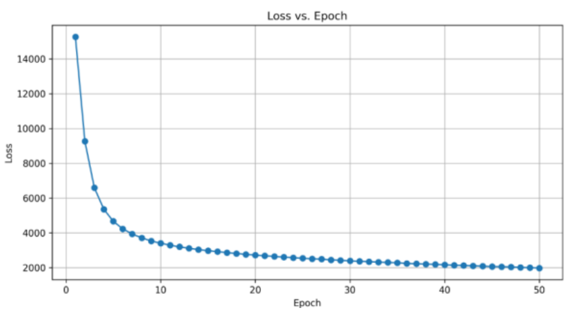
CPU Native



CPU BLAS



GPU Native



GPU cuBLAS